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In [1]: # Program to calculate the area of a rectangle
length = float(input("Enter the length of the rectangle: "))
width = float(input("Enter the width of the rectangle: "))
area = length * width
print(f"The area of the rectangle is: {area}")
```

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Enter the length of the rectangle: 15
Enter the width of the rectangle: 5
The area of the rectangle is: 75.0
```

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In [3]: # Program to calculate simple interest

principal = float(input("Enter the principal amount: "))
rate = float(input("Enter the rate of interest (in %): "))
time = float(input("Enter the time period (in years): "))
simple_interest = (principal * rate * time) / 100
print(f"The Simple Interest is: {simple_interest}")
```

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Enter the principal amount: 1500
Enter the rate of interest (in %): 5.5
Enter the time period (in years): 1
The Simple Interest is: 82.5
```

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In [4]: # Program to convert Celsius to Fahrenheit
celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = (celsius * 9 / 5) + 32
print(f"{celsius}°C is equal to {fahrenheit}°F")
```

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Enter temperature in Celsius: 26
26.0°C is equal to 78.8°F
```

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In [5]: # Program to find the average of three numbers
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
num3 = float(input("Enter third number: "))

average = (num1 + num2 + num3) / 3
print(f"The average of the three numbers is: {average}")
```

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Enter first number: 15
Enter second number: 47
Enter third number: 60
The average of the three numbers is: 40.666666666666664
```

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In [6]: # Program to find the square and cube of a number
number = float(input("Enter a number: "))
square = number**2
cube = number**3
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print(f"Square of {number} is: {square}")
print(f"Cube of {number} is: {cube}")
```

Enter a number: 15
Square of 15.0 is: 225.0
Cube of 15.0 is: 3375.0

In []:

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# Program to swap two numbers without using a temporary variable
a = float(input("Enter first number (a): "))
b = float(input("Enter second number (b): "))

print(f"\nBefore swapping: a = {a}, b = {b}")
a, b = b, a

print(f"After swapping: a = {a}, b = {b}")
```

In [8]:

```
# Student Report Card Program
# This program takes student details, marks for 5 subjects,
# and calculates the total and percentage.
student_name = input("Enter student name: ")
print("\nEnter marks for 5 subjects (out of 100):")
subject1 = float(input("Subject 1: "))
subject2 = float(input("Subject 2: "))
subject3 = float(input("Subject 3: "))
subject4 = float(input("Subject 4: "))
subject5 = float(input("Subject 5: "))
total_max_marks = 500
total_obtained = subject1 + subject2 + subject3 + subject4 + subject5
percentage = (total_obtained / total_max_marks) * 100
print("\n" + "=" * 30)
print(f"          STUDENT REPORT CARD          ")
print("'" + "=" * 30)
print(f"Student Name: {student_name}")
print("-" * 30)
print(f"Total Marks Obtained: {total_obtained} / {total_max_marks}")
print(f"Percentage Obtained : {percentage:.2f}%")
print("=" * 30)
```

Enter student name: Akash

Enter marks for 5 subjects (out of 100):
Subject 1: 75
Subject 2: 88
Subject 3: 84
Subject 4: 69
Subject 5: 91

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=====
          STUDENT REPORT CARD
=====
Student Name: Akash
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Total Marks Obtained: 407.0 / 500
```

Percentage Obtained : 81.40%
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